Topic 33 — Biosphere Reserves (जैवमंडल आरक्षित क्षेत्र)

Covers syllabus (click to expand)

Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र) | Major Biosphere Reserves of India | Biosphere Reserves of Uttar Pradesh (उत्तर प्रदेश) | Biosphere Reserves of the World

Sources baked in: UNESCO MAB; MoEFCC; NWDC; NCERT; UPPCS 2021–2025 PYQs **Weight:** ★★★ High — BR ↔ state; Nokrek ≠ Manipur; UP = zero BR **Last verified:** August 2026

Current Affairs (this topic)

- WNBR new listings drift the UNESCO count; India national BR list stays the teaching spine.
- Biosphere ≠ WPA statutory category.

33.1 Biosphere Reserve

Identity: UNESCO MAB 1971 | Core–Buffer–Transition | MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय) executive | No WPA category

- A **Biosphere Reserve (BR)** (जैवमंडल आरक्षित क्षेत्र) is a large landscape that conserves biodiversity while allowing regulated sustainable community use and research.
- UNESCO launched the **Man and the Biosphere (MAB) Programme in 1971** (same year as Ramsar (रामसर)).
- India notifies BRs through **MoEFCC executive action** — BRs are **not** a separate WPA 1972 statutory category; cores usually overlap an existing NP/sanctuary.
- **Three zones: Core** (no exploitative use / genetic diversity (आनुवंशिक विविधता)) → **Buffer** (research, education, regulated eco-tourism and grazing) → **Transition** (settlements, crop cultivation, livelihoods).
- **Triple mandate:** conservation + sustainable development + logistic support (research/monitoring/education).
- Edward Suess is classically linked to the **biosphere** concept; **biodiversity** as a contracted term is linked to **W.G. Rosen (1985)** after **Raymond F. Dasmann** used “biological diversity” (1968).
- Among common Prelims options, a **biosphere reserve** (जैवमंडल आरक्षित क्षेत्र) is the strongest answer for conserving biodiversity **along with traditional human life and cultural diversity**.
- India’s first BR is **Nilgiri (1986)** — Tamil Nadu, Karnataka, Kerala.
- India has **18 notified BRs**; about **12–13** are in UNESCO’s **World Network of Biosphere Reserves (WNBR)**.
- **Agasthyamalai** includes Neyyar, Peppara, Shendurney WLS and **Kalakad–Mundanthurai Tiger Reserve**.

- **Gulf of Mannar** is often the largest **UNESCO-certified** Indian BR by area; **Great Rann of Kutch** is the largest **national** BR.
- **Bundala** Biosphere Reserve is in **Sri Lanka**, not India.
- Area decreasing order among common options: **Nanda Devi > Pachmarhi > Agasthyamalai > Panna > Khangchendzonga**.
- Funding pattern under the Centrally Sponsored Scheme is typically **60:40** (Centre–State), **90:10** for NE/Himalayan states.

PYQ — UPPCS Prelims 2018, Q54

Which is NOT an example of in-situ (स्थान पर) conservation (स्व-स्थाने संरक्षण)?

Options: A. Biosphere reserve B. Botanical garden (वनस्पति उद्यान) C. National Park D. Sacred groves (पवित्र उपवन)

Show answer

Ans: B — Botanical garden is **ex-situ** (स्थानांतरित). BR / NP / sacred groves are in-situ.

> **Logic:** "Biosphere Reserves allow only absolute (निरपेक्ष) ban on all human use" is **false** — buffer/transition allow regulated use.

33.2 Major Biosphere Reserves of India

Identity: 18 national BRs | Match-list heavy | Nokrek / Manas / Simlipal traps

- **Largest BR:Great Rann of Kutch**, Gujarat. **Smallest:Nokrek**, Meghalaya.
- **Newest notified:Panna**, Madhya Pradesh (2011).
- **Manas (Assam)** remains a **national BR** but was **removed from UNESCO WNBR in 2011**.
- **Madhya Pradesh** has three BRs: Pachmarhi, Achanakmar–Amarkantak (with Chhattisgarh), Panna.
- Multi-state classics: **Nilgiri** (TN–KA–KL), **Agasthyamalai** (KL–TN), **Achanakmar–Amarkantak** (MP–CG).

PYQ — UPPCS Prelims 2025, Q144

Which pair(s) NOT correctly matched? 1. Nokrek — Manipur 2. Manas — Assam 3. Agasthyamalai — Kerala

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

Show answer

Ans: D — Only stmt 1 is wrong. Nokrek = **Meghalaya**.

PYQ — UPPCS Prelims 2024, Q10

Match: A. Simlipal B. Nokrek C. Agasthyamalai D. Manas with 1. Kerala 2. Odisha 3. Assam 4. Meghalaya

Options: A. 1 2 4 3 B. 3 1 4 2 C. 2 4 3 1 D. 2 4 1 3

Show answer

Ans: D — Simlipal=Odisha; Nokrek=Meghalaya; Agasthyamalai=Kerala; Manas=Assam.

PYQ — UPPCS Prelims 2021, Q140

Match: A. Similipal B. Pachmarhi C. Nandadevi D. Nokrek with 1. Uttarakhand 2. Meghalaya 3. Madhya Pradesh 4. Odisha

Options: A. 4 3 2 1 B. 4 3 1 2 C. 1 3 4 2 D. 3 4 1 2

Show answer

Ans: B — Simlipal=Odisha; Pachmarhi=MP; Nanda Devi=Uttarakhand; Nokrek=Meghalaya.

PYQ — UPPCS Prelims 2022, Q110

Match: A. Manas B. Sundarbans C. Nanda Devi D. Pachmarhi with 1. Madhya Pradesh 2. Uttar Pradesh 3. Assam 4. West Bengal

Show answer

Intended Ans: D (A-3, B-4, C-2, D-1) — paper's List-II labelled Nanda Devi as "Uttar Pradesh."
Geography fact: Nanda Devi BR is in **Uttarakhand**. UP itself has **zero** BRs.

All 18 Biosphere Reserves of India

#	Biosphere Reserve	State(s)	Notified	WNBR note
1	Nilgiri	TN, KA, KL	**1986**	First; WNBR 2000
2	Nanda Devi	**Uttarakhand**	1988	WNBR 2004
3	Nokrek	**Meghalaya**	1988	Smallest; WNBR 2009
4	Gulf of Mannar	Tamil Nadu	1989	Marine; WNBR 2001
5	Sundarbans	West Bengal	1989	WNBR 2001
6	Manas	Assam	1989	**Removed UNESCO 2011**
7	Great Nicobar	A&N	1989	WNBR 2013
8	Simlipal	**Odisha** (ओडिशा)	1994	WNBR 2009
9	Dibru-Saikhowa	Assam	1997	National only
10	Dihang-Dibang	Arunachal Pradesh	1998	National only
11	Pachmarhi	Madhya Pradesh	1999	WNBR 2009
12	Khangchendzonga	Sikkim	2000	WNBR 2018

#	Biosphere Reserve	State(s)	Notified	WNBR note
13	Agasthyamalai	Kerala, Tamil Nadu	2001	WNBR 2016
14	Achanakmar–Amarkantak	MP, Chhattisgarh	2005	WNBR 2012
15	Great Rann of Kutch	Gujarat	2008	**Largest**
16	Cold Desert	Himachal Pradesh	2009	Recent WNBR addition
17	Seshachalam Hills	Andhra Pradesh	2010	National only
18	Panna	Madhya Pradesh	**2011**	Newest; WNBR 2020

> **Logic: Agasthyamalai** spans **Kerala and Tamil Nadu** — pairing it with Kerala alone is still accepted when Tamil Nadu is not an option.

33.3 Biosphere Reserves of Uttar Pradesh

Identity: Zero BRs | Nearest = Nanda Devi (UK) | Dudhwa ≠ BR

- **Uttar Pradesh has no notified Biosphere Reserve.**
- Nearest Himalayan BR is **Nanda Devi (Uttarakhand)** — not in UP after the 2000 bifurcation.
- Do not treat **Dudhwa NP**, **Pilibhit TR**, or **Valmiki NP (Bihar)** as UP biosphere reserves.
- Terai (तराई) Arc (Dudhwa–Kishanpur–Katarniaghat) is landscape-conservation geography, but it is **not** a notified BR.
- Neighbour contrast: MP has **three** BRs; Bihar and Rajasthan also have **zero** BRs (Valmiki is an NP).

State	BR count	Fact
Uttar Pradesh (उत्तर प्रदेश)	**0**	Any “UP BR” option is false
Uttarakhand	1	Nanda Devi
Madhya Pradesh	3	Pachmarhi, Achanakmar–Amarkantak, Panna
Bihar	0	Valmiki = NP, not BR

> **Logic:** “Which BR is in UP?” → **None**.

33.4 Biosphere Reserves of the World

Identity: UNESCO WNBR | Global MAB sites | Living laboratories

- UNESCO’s **World Network of Biosphere Reserves** links hundreds of sites worldwide under MAB for conservation and sustainable development.

- Famous icons: **Yellowstone (USA)**, **Serengeti–Ngorongoro (Tanzania)**, **Galápagos (Ecuador)**, **Lake Baikal (Russia)**, **Doñana (Spain)**, **Mount Kenya (Kenya)**.
- World BRs follow the same **core–buffer–transition** design as India's national BRs.
- A site can carry **BR + National Park + World Heritage** together (e.g., Sundarbans-type landscapes).
- Periodic review (~10 years) can lead to **delisting** from WNBR (India example: **Manas 2011**).

World BR / landscape	Country	Fact
Yellowstone	USA	Iconic (साकार) MAB / NP landscape
Serengeti–Ngorongoro	Tanzania	Savanna (सवाना) mega-fauna
Galápagos	Ecuador	Evolution / WH overlap
Lake Baikal	Russia	Freshwater biodiversity
Doñana	Spain	Wetland BR classic
Mount Kenya	Kenya	Afro-alpine BR

> **Logic:** World BR network ≠ World Heritage list — related but different UNESCO instruments.
